Project Design Phase-II
Technology Stack (Architecture & Stack)

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Project Name	Strategic product placement analysis: unveiling sales impact with tableau visualization

Technical Architecture:

Data Sources:
Integrates unstructured planograms/calendars from Cloud Storage (Google Drive) with structured POS/inventory data from Database Servers.

Preparation:
Uses Tableau to clean SKU inconsistencies and join "Placement" metadata with actual "Sales" figures.

Analytics Engine:
A core modeling layer that calculates specific KPIs like Eye-Level Lift and End-Cap Conversion Rates.

Distribution:
Hosted on Tableau Server/Cloud for security and automated data refreshes.

Accessibility:
Features a Responsive Design optimized for mobile, tablet, and desktop viewing.

User Action:
Enables Category Managers to make rapid "Keep or Move" decisions based on real-time impact visualization.

Core Impact Metrics

To keep the analysis focused, the system prioritizes these three high-level views:

Placement ROI:
Sales growth vs. the cost of premium shelf space.

Heatmap Analysis:
Visualizing "Hot Zones" vs. "Dead Zones" within a store aisle.

A/B Testing:
Direct comparison of sales performance across two different placement strategies

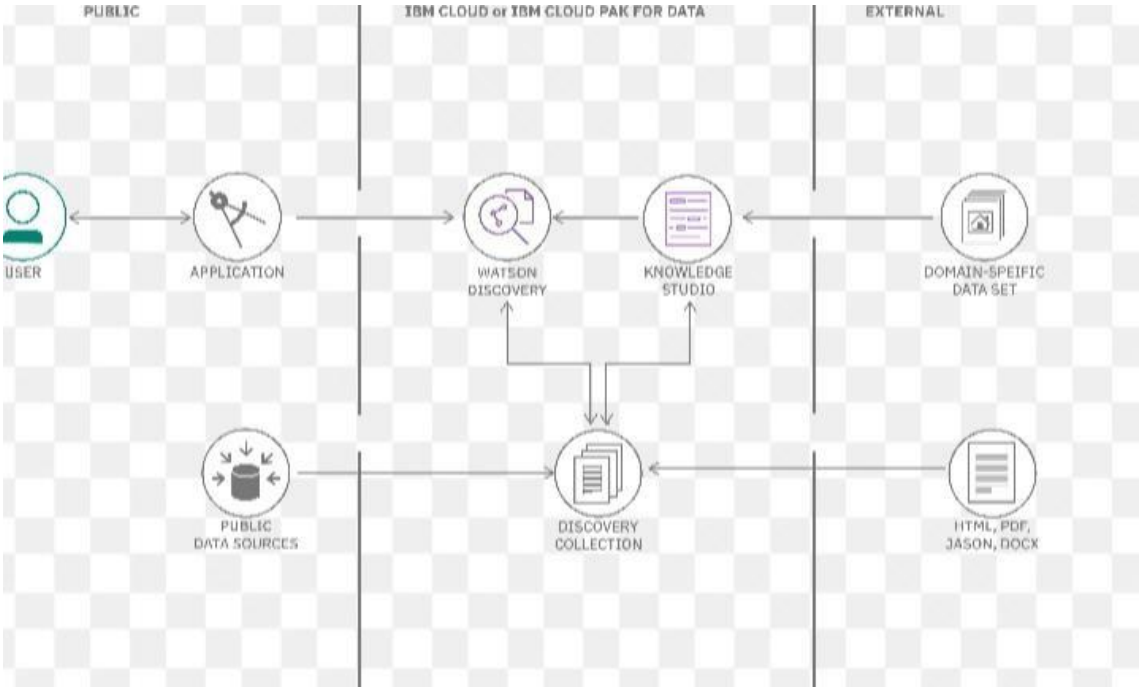


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Responsive Web and Mobile UI for interactive dashboard viewing.	Tableau Web / Mobile App
2.	Application Logic-1	Data harmonization and ETL to merge planograms with sales data.	Tableau Prep / Python
3.	Application Logic-2	Impact modeling to calculate Eye-level vs.	Bottom-shelf sales lift. Tableau Desktop / SQL
4.	Application Logic-3	Automated refresh scheduling for real-time sales trend	Tableau Server Scheduler

		monitoring.	
5.	Database	Stores structured historical POS transactions and store metadata.	MySQL / PostgreSQL
6.	Cloud Database	High-speed warehouse for aggregating multi-store retail datasets.	Snowflake / Google BigQuery
7.	File Storage	Storage for flat-file planograms and manual store audit logs.	Google Drive / Local Filesystem
8.	External API-1	Normalizes sales data based on local weather or seasonal trends.	IBM Weather API
9.	External API-2	Automates user permission syncing and programmatic report delivery.	Tableau REST API
10.	Machine Learning	Forecasts potential revenue lift for proposed shelf-space changes.	Python (Scikit-learn)
11.	Infrastructure	Secure cloud deployment for 24/7 multi-device accessibility.	Tableau Cloud / Kubernetes

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Utilizing open-source tools for data transformation and	Python (Pandas/NumPy), PostgreSQL, Open-source API

		visualization agility.	connectors
2.	Security Implementations	Multi-layered security with Row-Level Security (RLS) and encrypted data transit.	SHA-256, SAML/SSO, Tableau RLS, SSL/TLS Encryption.
3.	Scalable Architecture	A 3-tier architecture (Data, Logic, Presentation) designed for multi-device growth.	Micro-services, Kubernetes, Tableau Cloud Clusters.
4.	Availability	High availability through distributed servers and automated health checks.	Load Balancers, AWS Multi-AZ, Tableau Server High Availability.
5	Performance	Optimized query execution through data extracts and caching for fast load times.	Redis Caching, Tableau Data Hyper Extracts, CDN (CloudFront).